

Dementia Risk Prediction with the SurvivEHR Foundation Model on UK Biobank Linked Primary Care + Hospital Data

Progress Report

Cohort: UK Biobank linked GP primary-care + HES hospital records (n = 133,322) **Report date:** 2026-05-15

1. Executive Summary

We are building a competing-risk dementia prediction model on UK Biobank-linked **GP primary-care records** (Read v2 / CTV3, ~133K cohort patients) and **HES hospital records** (ICD-10). The model uses the **SurvivEHR architecture** (Gadd et al., 2025, medRxiv) — a GPT-style transformer foundation-model for EHR time-to-event prediction — pretrained on our local UKB GP records and fine-tuned for dementia. It outputs a full **25-year cumulative incidence function (CIF)** for both dementia and competing death via a DeSurv ODE head.

The two strongest results on the canonical 8,241-patient test cohort are summarised below. Neither model wins every metric: **Single-GP + HESstatic** is best on top-1% PPV (clinical screening yield) and on dementia IBS calibration; **Dual + HESstatic + SST1%** is best on time-dependent C-index (overall discrimination) and on broader-cut PPV (top 5%).

Metric (5-year horizon)	Single-GP + HESstatic (<i>clean baseline; best top-1% PPV, best IBS dem</i>)	Dual (GP + HES) + HESstatic + 1% SST (<i>best C-index, best PPV@5%</i>)
Time-dependent C-index (Antolini), dementia	0.8451	0.8506
C-index, death	0.9622	0.9589
Integrated Brier Score, dementia (lower = better)	0.3152	0.3265
5-year AUROC (binary, exclude_early_censored)	0.9236	0.9271
5-year PPV @ top 1% / 5% / 10%	68.9% / 27.9% / 19.0%	55.6% / 29.7% / 19.3%

For external context, the most directly comparable foundation-model EHR paper, **TRADE** (Wisnivesky et al., NYU AD/ADRD/MCI prediction), reports 5y AUROC = 0.735 and PPV@top 1% = 39.2%. Detailed code-set overlap analysis (§8) confirms our HES dementia label is a near-strict subset of TRADE's umbrella, with the main gaps being MCI (no UK ICD-10 equivalent) and FTD/Lewy body sub-codes (~160 extra patients available if we extend our label).

2. Data Sources

2.1 The two data sources

We use **two UK Biobank-linked sources**:

1. **UKB-linked GP primary-care extract** — General Practice records (Read v2 / CTV3 clinical coding) covering 502,246 UKB participants in total (of which 230,001 have ≥ 1 diagnostic record). Contains the full longitudinal record of GP encounters: diagnoses, prescriptions, lab tests, lifestyle observations, referrals, and administrative events.
2. **UKB-linked HES (Hospital Episode Statistics)** — Secondary-care records (ICD-10 diagnoses, OPCS-4 procedures) covering 448,985 UKB participants. Contains admission dates, primary/secondary diagnosis lists, and operation lists.

The two sources are linked at the patient level via UKB's encrypted participant ID.

2.2 GP primary-care record is NOT just diagnoses

A common misunderstanding is that GP records contain only diagnosis codes. In reality, the Read v2 vocabulary used in UK GP systems contains **108,118 distinct tokens** in our pretrained model's vocabulary, spanning a wide range of clinical event types.

Read v2 uses semantic prefixes to group code categories. The main chapters are roughly:

Chapter prefix	Category
1 . . .	History and process codes (symptoms, presenting complaint)
2 . . .	Examination findings, physical measurements (BP, weight, BMI)
3 . . .	Diagnostic procedures (ECG, imaging)
4 . . .	Laboratory tests (blood, urine, etc.)
6 . . .	Administrative / preventive (screening, counselling)
8 . . .	Other procedures (vaccinations, minor surgery)
9 . . .	Administrative (letters, referrals, registrations)
B . . .	Neoplasms
E . . .	Mental disorders
F . . .	Nervous system (incl. dementia)
G . . .	Circulatory (cardiovascular)
H . . .	Respiratory
J . . .	Digestive
K . . .	Genitourinary
Eu0 . .	ICD-10 mapped mental disorders (incl. dementia F00-F09)

So in the GP record stream one can see: a diagnostic code (e.g., G20 . . essential hypertension), then a blood test result (44 . . cholesterol), then a prescription, then a referral letter (9 . .), then a measurement (229 . . body weight), etc. The model sees this **chronologically ordered stream** of event codes + dates + numeric values (where applicable), **not** an aggregated feature vector.

Example patient (anonymised, real record from our test cohort):

PATIENT_ID: 2442387
YEAR_OF_BIRTH: 1941 SEX: M IMD: 3 ETHNICITY: White British
INDEX_DATE: 2013-01-01 (= 72nd birthday)

Total GP events recorded: 379 events from 1961-01-01 to 2012-12-31
HES pre-index admissions: 4 (HES_TOTAL_ADMISSIONS scaled = 0.279)
HES comorbidity flags: none of stroke/MI/HF/HT/diabetes etc. se

First 10 GP events (chronological), raw codes:

1961-01-01 J16y4
1971-01-01 H51..
1974-01-01 G20..
2000-01-01 1241..
2000-01-01 12C1..
2000-01-01 12C3..
2000-01-01 12A1..
2000-01-01 12C4..
2002-12-01 G5y34
2005-08-01 J51..

... (total 379 events; final event determines the patient's outcome lab

The descriptive labels for each Read v2 code are NOT used by the model — the model only sees the raw code tokens. Translation of any specific code to a description requires a Read v2 dictionary lookup (we have not embedded such a dictionary in the report, as the model does not rely on it).

2.3 Cohort composition

Stage	Patient count
Total UKB-linked GP primary-care database (static_table, all UKB participants with GP linkage)	502,246
Of these, with ≥ 1 diagnostic record in diagnosis_table	230,001
Total UKB-linked HES hospital database	448,985
In BOTH GP and HES (HES is a subset of the GP-linked set)	448,985
GP-only (no HES linkage)	53,261
Meeting study inclusion (≥ 1 y GP registration, ≥ 5 events before idx 72, idx age in 50-90)	133,322
After GP-prevalent dementia leakage removal (cleanest cohort)	133,076

Dementia capture by data source (in the full UKB linkage layer, before applying our cohort filter):

Source	Dementia patients	Note
GP (31 strict Read v2 dementia codes)	992	Formal GP-coded dementia
HES (F00., F01., F02., F03, G30.)	8,941	Hospital ICD-10 dementia

In BOTH GP and HES	781	
GP-only	211	
HES-only	8,160	HES catches $\sim 9\times$ more dementia than GP
Total unique (GP $\cup$ HES)	9,152	

Key finding: HES catches **almost an order of magnitude more dementia patients than GP**. UK GP records substantially under-code dementia — patients are often diagnosed in memory clinics or hospital admissions (which goes to HES) but the GP record is never updated. This is consistent with the well-documented 30–50% GP under-diagnosis rate (Lang 2017; Connolly 2011) and strongly motivates the self-training pipeline.

Within our 133,322 study cohort (subset of the 502,246 full linkage after applying inclusion criteria: idx age 72, ≥ 1 y GP registration, ≥ 5 events before index):

Has data in	Patients (% cohort)
GP records (all 133,322 must have GP — inclusion criterion)	133,322 (100%)
With ≥ 1 valid ICD-10 HES record (any time)	126,530 (94.9%)
GP-only dementia (31 strict, not in HES)	162 (0.12%)
HES-only dementia (not in GP)	5,477 (4.11%)
Both GP and HES dementia	519 (0.39%)
Total dementia patients (GP $\cup$ HES, ever)	6,158 (4.62%)

The 6,158 within-cohort number is smaller than the 9,152 full-linkage number because **the inclusion criteria drop 2,994 dementia patients** who fail one or more criteria (e.g., do not have a GP visit at age 72, insufficient GP registration history, or fewer than 5 pre-index events). The within-cohort prevalence (4.62%) is approximately $2\times$ the UK general elderly baseline because we deliberately recruited at age 72.

2.4 Train / Val / Test split

Splits are made at the **GP practice level** (PRACTICE_ID), not at the patient level. All patients in a given practice go to the same split. This prevents practice-level data leakage (e.g., regional coding patterns, practice-specific terminology).

Split	Patients	Dementia events	Death events	Censored
Train	119,271	5,286	7,898	106,087
Val	5,794	296	485	5,013
Test	8,257	376 (370 after canonical filter to 8,241)	451	7,430
Total	133,322	5,958	8,834	118,530

Dementia event prevalence across the full cohort = $5,958 / 133,322 = 4.47\%$; death prevalence = 6.63% ; censored = 88.91% . Splits are approximately stratified by event rate.

(Note: 5,958 incident dementia events vs 6,158 patients with any dementia code in GP/HES history — see §2.3 — differ by ~ 200 . The gap is patients whose dementia code occurs before the prediction point, or who die from another cause first; these are not labeled as "incident dementia event" in the V2 dataset.)

△□ **Important caveat about train/val/test cleanliness:** The 246 GP-prevalent dementia patients (213 train / 17 val / 16 test) — i.e., patients whose dementia code appears **before** the prediction point — are present in the V2 dataset and were used to train all five "V2 clean labels" post-leakage models (Single-GP + HESstatic through Dual + HESstatic + SST5%). **Only the test set is filtered** to the canonical $8,241 = 8,257 - 16$ for evaluation. The "leakage-fixed" base dataset that removes all 246 patients is used by the most recent model **Single-GP + HESstatic + leak-fix** (§5, §7.1). All test-cohort metrics in §7.1 are therefore on the same canonical 8,241 patients; training-set leakage of $213 + 17 = 230$ patients (0.18% of train + val) was tolerated in the V2-labelled comparison set to keep model comparisons consistent with our historical experimental record.

3. Label Definition

3.1 Index date and prediction setup (important clarification)

The study fixes **age 72 as the cohort entry date** for inclusion (all patients must have at least one GP visit at age 72), but this is **not the prediction time-point**. The model performs **dynamic prediction**:

```
Past:  [..... GP events .....]
                                     ↑
                                     Prediction point = the patient's most recent
                                     pre-index GP encounter (often months/years BE
```

Future: [...] → dementia / death / censored
time = days from prediction point (NOT from age 72)

The model emits CIF over $t = 0 \rightarrow 25$ years from the prediction point. This matches how the model is actually deployed: a clinician asks "given everything I know about this patient up to today, what is their risk over the next N years?" — not "what is their risk from a fictitious cohort entry date."

The "index age 72" is used only for **inclusion / exclusion**, not for time origin. Events occurring before the prediction point are model input; events occurring after are target outcomes.

3.2 Dementia label: GP side (41 Read v2 codes)

The full list of Read v2 / CTV3 codes used to identify dementia events on the GP side, with **patient counts in the raw GP database (230,001 patients with diagnostic records, drawn from the 502,246 UKB-linked GP cohort)** — note: these are unique-patient counts, not event counts.

Read code	Patients	Description
F110.	449	Alzheimer's disease (neurology chapter)
Eu00.	220	Dementia in Alzheimer's disease (ICD-10 F00 umbrella)
Eu01.	78	Vascular dementia (F01 umbrella)
Eu02z	70	Unspecified dementia (F02.9 default)

Eu002	60	Dementia in AD, atypical or mixed type (F00.2)
Eu057	57	Mild cognitive disorder — closest available to MCI (F06.7)
E00..	54	Senile/presenile organic psychotic conditions (umbrella)
Eu023	49	Dementia in Parkinson's disease (F02.3)
Eu00z	45	Dementia in AD, unspecified (F00.9)
Eu04.	17	Delirium (F05) — broader organic
Eu025	15	Dementia in other specified diseases (F02.5)
Eu01z	13	Vascular dementia, unspecified (F01.9)
E001.	13	Presenile dementia
F1100	12	Alzheimer's, early onset
Eu001	10	Dementia in AD, late onset (F00.1)
E004.	8	Arteriosclerotic dementia
Eu053	8	Organic mood disorders (F06.3)
Eu000	7	Dementia in AD, early onset (F00.0)
Eu04z	7	Delirium, unspecified
Eu02.	6	Dementia in other diseases (F02 umbrella)
Eu013	5	Mixed cortical/subcortical vascular dementia (F01.3)
E000.	5	Uncomplicated senile dementia
Eu0z.	5	Unspecified organic mental disorder
Eu060	5	Organic personality disorder (F07.0)
Eu054	5	Organic anxiety disorder (F06.4)
Eu05y	4	Other organic mental disorders
Eu052	4	Organic delusional disorder
Eu01y	3	Other vascular dementia
E001z	3	Presenile dementia NOS
Eu062	3	Postconcussional syndrome (F07.2)
F1101, Eu020 (Pick's / FTD), E004z, E0021	2 each	rare codes
Eu02y, Eu012, Eu011, E00z., E0040, E003., E0020	1 each	very rare

Union of any of 31 strict dementia codes in the 230K GP DB: 992 patients (0.43%)

Union of all 41 codes (incl. 10 broader organic): 1,077 patients (0.47%)

Within our 133K study cohort: 681 (31-strict) / 722 (all 41) patients — i.e., only ~0.5% of cohort has any GP-coded dementia.

Notes:

- UK GPs overwhelmingly use F110. (Alzheimer's) and the Eu0. . family, with most

patients coded under the unspecified subcategory (Eu02z, Eu00z). Fine-grained subtype coding (specific vascular subtypes, etc.) is rare.

- The "broader organic" 10 codes (Eu04, Eu05y, Eu057, Eu060, etc.) add only ~80 patients beyond the 31 strict codes.
- **GP-coded dementia is rare in this cohort** because of well-documented GP under-coding: many patients are diagnosed in memory clinics / hospitals (going to HES) but the GP record is never updated. See §2.3 — HES catches $\sim 9\times$ more dementia.

3.3 Dementia label: HES side (ICD-10)

HES uses WHO ICD-10. We use the standard "broad dementia" definition:

ICD-10 family	Patients	%cohort	Description
F00.* (F00.0, F00.1, F00.2, F00.9)	2,288	1.72%	Dementia in Alzheimer's disease
F01.* (F01.0, F01.1, F01.2, F01.3, F01.8, F01.9)	1,417	1.06%	Vascular dementia
F02.* (F02.0, F02.1, F02.2, F02.3, F02.8)	708	0.53%	Dementia in other diseases (Pick's, Huntington's, CJD, Parkinson's, etc.)
F03	3,048	2.29%	Unspecified dementia
G30.* (G30.0, G30.1, G30.8, G30.9)	2,817	2.11%	Alzheimer's disease

Union of any HES dementia code: 5,996 patients (4.50% of cohort)

Combined GP + HES dementia label in the cohort:

	n	%cohort
GP-only (31-strict GP code, no HES dementia code)	162	0.12%
HES-only (HES dementia code, no GP code)	5,477	4.11%
Both GP and HES	519	0.39%
Union (any source)	6,158	4.62%

Within the test cohort ($n=8,241$), after restricting to events occurring **after** the prediction point and before death, this yields **370 incident dementia events** (i.e., true "future dementia" labels for evaluation).

3.4 22-dim HES static covariates (handcrafted features)

In addition to feeding GP event sequences into the transformer, we compute a 22-dimensional patient-level summary vector from pre-index HES records (strict temporal filter — `admidate < index_date`). Dementia codes (F00-F03, G30) are **excluded** from these features to prevent label leakage.

The 22 features:

#	Feature	Type	ICD-10 prefix
0	HES_TOTAL_ADMISSIONS	log-scaled count	
1	HES_TOTAL_UNIQUE_DIAG	log-scaled count	

2	HES_HAS_STROKE	binary	I60-I69
3	HES_HAS_MI	binary	I21-I22
4	HES_HAS_HEART_FAILURE	binary	I50
5	HES_HAS_DIABETES	binary	E10-E14
6	HES_HAS_DELIRIUM	binary	F05
7	HES_HAS_TBI	binary	S06
8	HES_HAS_HYPERTENSION	binary	I10-I15
9	HES_HAS_ATRIAL_FIBRILLATION	binary	I48
10	HES_HAS_CKD	binary	N18
11	HES_HAS_DEPRESSION	binary	F32, F33
12	HES_HAS_PARKINSON	binary	G20
13	HES_HAS_EPILEPSY	binary	G40, G41
14	HES_HAS_OBESITY	binary	E66
15	HES_HAS_HYPERLIPIDEMIA	binary	E78
16	HES_HAS_COPD	binary	J44
17	HES_HAS_ALCOHOL	binary	F10
18	HES_HAS_SLEEP_DISORDER	binary	G47
19	HES_MEAN_STAY_DAYS	log-scaled continuous	
20	HES_EMERGENCY_RATIO	continuous [0,1]	admimeth 21-28
21	HES_YEARS_SINCE_LAST_ADMISSION	scaled continuous	

These 22 dimensions are concatenated with 27 baseline demographic dimensions (SEX one-hot, IMD one-hot, ETHNICITY one-hot, YEAR_OF_BIRTH scaled) to form a **49-dim static covariate vector** that is projected to the model's embedding space and **added to every token embedding** in the sequence.

4. Model Architecture

4.1 Pretraining (done locally on UKB data, frozen backbones)

We use the **SurvivEHR architecture** (Gadd et al., 2025) and pretrain it from scratch on our own UKB-linked data. We do **not** use the SurvivEHR authors' public CPRD-pretrained weights.

GP backbone pretrain (crPreTrain_small_1337.ckpt):

- Self-supervised next-event prediction with competing-risk time-to-event objective
- Data: full UKB-linked GP `example_exercise_database.db` ($\approx 502K$ patients in `static_table`, $\sim 230K$ with diagnostic records)
- Vocabulary: 108,118 tokens (Read v2 / CTV3)
- Architecture: 6 transformer layers, 6 heads, 384 embedding dim
- Block size: **256** at pretrain (matches `config_CompetingRisk11M.yaml`); the same checkpoint is re-used at block size 512 in fine-tune (positional encoding generalises)
- Pretrain hyperparameters: batch 64, see config file for further details

HES backbone pretrain (crPreTrain_HES_1337.ckpt):

- 419,966 patient HES sequences, ~3.9M events total
- 1,499-token vocabulary (ICD-10, 3-character truncated)
- Same transformer architecture, block size 256
- Trained for 8 epochs with early-stopping (config target was 15, best checkpoint = `crPreTrain_HES_1337-epochepoch=07.ckpt`), batch 64

4.2 Fine-tuning architectures

Two backbone configurations:

Single-backbone (GP-only) fine-tune (used in **Single-GP** + **HESstatic**, the just-trained **Single-GP** + **HESstatic** + **leak-fix**, and the planned **SST1%** extension):

```
GP token sequence (≤512 tokens) → GP Backbone (pretrained, frozen-then-
→ patient embedding h_gp (384-dim)
49-dim static covariates (incl. 22 HES summary) → Linear(49→384) → stat
→ SUMMED with every position's token em
→ DeSurv ODE Competing-Risk Head
→ CIF_dementia(t), CIF_death(t) for t
```

Dual-backbone fine-tune (used in **Dual-gated** + **HESstatic** and its SST variants):

GP data $\rightarrow$ [GP Backbone, 108K vocab, block 512] $\rightarrow$ h_gp (384-dim) $\neg$
HES data $\rightarrow$ [HES Backbone, 1.5K vocab, block 256] $\rightarrow$ h_hes (384-dim) $\neg$ $\vdash$ Ga

Gated fusion:

```
gate =  $\sigma(W_g \cdot [h_{gp}; h_{hes}])$   
h_fused = gate  $\odot$  W_gp(h_gp) + (1-gate)  $\odot$  W_hes(h_hes)
```

Both backbones have their own 49-dim and 27-dim static covariate projections respectively. Total dual-model parameters: $\sim 106\text{M}$ trainable.

4.3 DeSurv competing-risk head

- Gaussian-mixture distribution over time
- Fixed-step ODE solver integrates the mixture into a cumulative incidence function
- Two cause-specific outputs satisfying $\text{CIF_dem}(t) + \text{CIF_death}(t) \leq 1$ by construction
- Time scaling: time_scale (days) = $1825 \times \text{supervised_time_scale}$ (5) = 9,125 days $\approx$ **25 years**
- Output grid: 1,000 evenly-spaced points on $t_{\text{eval}} \in [0, 1]$ (= 0 to 25 years)

4.4 Self-training (SST) — iterative 3-round procedure

To address the well-documented under-diagnosis of dementia in primary care (30–50% missing, Lang 2017; Connolly 2011), we apply **three iterative cumulative self-training rounds** on top of the base dual-backbone model. Each round uses the **previous round's trained model** for inference and adds new pseudo-positives on top of the previous round's pseudo set.

Round 0 (base): Train Dual-gated + HESstatic on V2 clean labels (real)

Round 1 (SST1%):

- Use Round-0 model to infer dementia CIF on the train
- Among censored patients, select top-1%-threshold = 7
- Sample pseudo event times from the empirical alive-c distribution ($n=3,637$, median 8.8 years, max ~16 yea
- Retrain on (real events $\cup$ 771 pseudo).
- Output: "Dual + HESstatic + SST1%" (= 771 pseudo tot

Round 2 (SST2%):

- Use Round-1 model to infer on training set.
- Add NEW 824 pseudo-positives (top-2% threshold), dis
- Retrain on (real events $\cup$ R1 pseudo $\cup$ R2 new pseudo)
- Output: "Dual + HESstatic + SST2%" (= 1,595 pseudo t

Round 3 (SST5%):

- Use Round-2 model to infer.
- Add NEW 2,219 pseudo-positives (top-5% threshold).
- Retrain on (real $\cup$ R1 $\cup$ R2 $\cup$ R3 new pseudo).
- Output: "Dual + HESstatic + SST5%" (= 3,814 pseudo t

For each pseudo-positive, the event time is set as `prediction_point + sampled_lag`, where the lag is drawn (with seed=1337) from the empirical alive-censored lag distribution. CIF is queried at the model's maximum supervised horizon (`t_eval[-1]`, normalised $\rightarrow$ 25 years) for ranking.

The just-trained "Single-GP + HESstatic + leak-fix" model (§7.1) is the leakage-cleaned base on top of which the single-round SST will be applied next (planned, §5 last row).

4.5 Training hyperparameters

	Value
Per-GPU batch size	16 (dual) / 32 (single)
Effective batch (after grad-accum)	$16 \times 32 = 512$ (dual), or $32 \times 16 = 512$ (single)
Learning rate (backbone)	5e-5
Learning rate (survival head)	5e-4 ($10 \times$ backbone)
Optimizer	AdamW
Scheduler	ReduceOnPlateau (factor 0.8)
Epochs	up to 25, early-stop patience 10
Seed	1337
Hardware	1 $\times$ NVIDIA GPU (24 GB), 48 CPUs

5. Experiments

Each experiment is identified below by its architecture configuration. Each name encodes whether the GP-only or dual backbone is used, whether HES static features are included, and whether self-training pseudo-labels are added.

Short name	Architecture	Label dataset	Self-training
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dementia within 5 years of the prediction point?"), AUROC = the probability the model ranks a random positive above a random negative.

- Range: [0, 1], 0.5 = random
- Computed by extracting `CIF_dementia` at the time-grid index closest to $t = 0.2$ (= 5 years on our 25-year normalised horizon $t_{eval} \in [0, 1]$), then running standard binary AUROC

Important — time origin: 5y is measured **from the prediction point** (the patient's most recent pre-index GP event), not from the fixed inclusion age 72. This matches our dynamic-prediction setup (§3.1). This differs from TRADE, where "5y" is measured from a sliding-window index visit — both are valid but the time origins are not identical when comparing absolute numbers.

Censoring policy: We exclude patients censored before 5y (their 5y outcome is unknown), as used in fixed-horizon EHR benchmarks like TRADE. Patients who died within 5y without dementia count as 5y negatives.

6.4 PPV @ top X% (positive predictive value at top X% screening cut-off)

What it measures: Of the X% highest-risk patients flagged by the model, what fraction actually developed the event by the horizon. Directly maps to **clinical screening yield**.

- Range: [0, 1], depends on prevalence
- **Baseline (random screen) = prevalence**
- We report PPV at top 1%, 5%, 10%, and report **lift** = **PPV / prevalence** for cross-cohort comparison

7. Results

7.1 Cohort-level full results (post-leakage clean models)

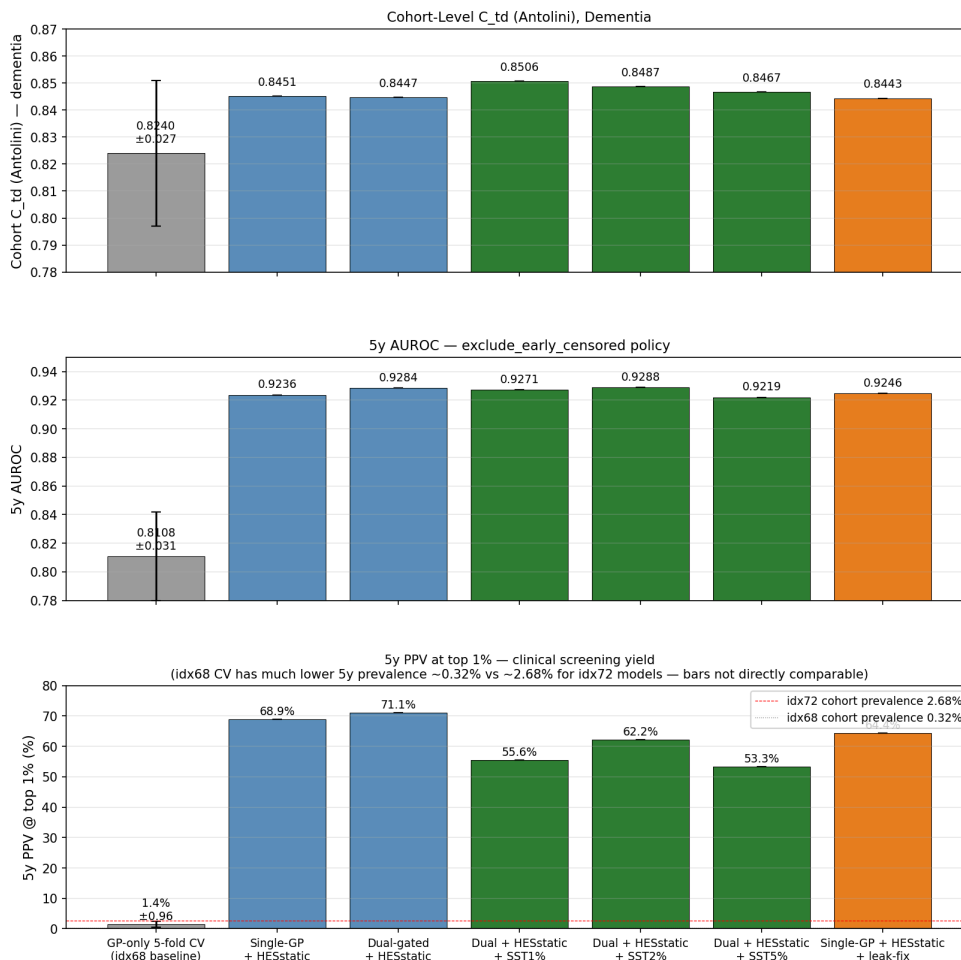
All metrics computed on the canonical 8,241-patient test cohort with V2 clean labels.

Model (architecture)	Dementia C_td	Death C_td	Overall C_td	IBS dem	IBS death	5y AUROC	PPV@1%	PPV@5%	PPV@10%
GP-only 5-fold CV (<i>idx</i> age 68 — baseline, different cohort)	0.8240 ± 0.027	—	—	—	—	0.8108 ± 0.031	1.43% ± 0.96%	1.22% ± 0.31%	0.93% ± 0.27%
Single-GP + HESstatic	0.8451	0.9622	0.9017	0.3152	0.0932	0.9236	68.89%	27.88%	19.03%
Dual-gated + HESstatic	0.8447	0.9582	0.9005	0.3395	0.0805	0.9284	71.11%	27.88%	19.25%
Dual + HESstatic + SST1%	0.8506	0.9589	0.9038	0.3265	0.1241	0.9271	55.56%	29.65%	19.25%

Dual + HESstatic + SST2%	0.8487	0.9590	0.9017	0.2773	0.1972	0.9288	62.22%	28.76%	18.81%
Dual + HESstatic + SST5%	0.8467	0.9603	0.9066	0.2713	0.3194	0.9219	53.33%	30.97%	18.58%
Single-GP + HESstatic + leak-fix (just finished — leakage-cleaned V6 base, no SST yet)	0.8443	0.9599	0.8962	0.2887	0.1153	0.9246	64.44%	30.09%	18.81%

Caveat on the idx68 GP-only 5-fold CV baseline: This row uses a **different cohort** (index age 68 instead of 72, single-GP-backbone, no HES static, no SST). 5-year prevalence in its evaluable subset is $\sim 0.32\%$ (vs 2.68% for the idx72 cohort), so its PPV @ top X% is not directly comparable to the other rows. C_{td} and AUROC are more comparable but still see a different prediction task. The idx72 GP-only 5-fold CV (which would be directly comparable) is currently in progress and will replace this row when finished. A pending idx = 72 5-fold CV replacement is planned in §10.1.

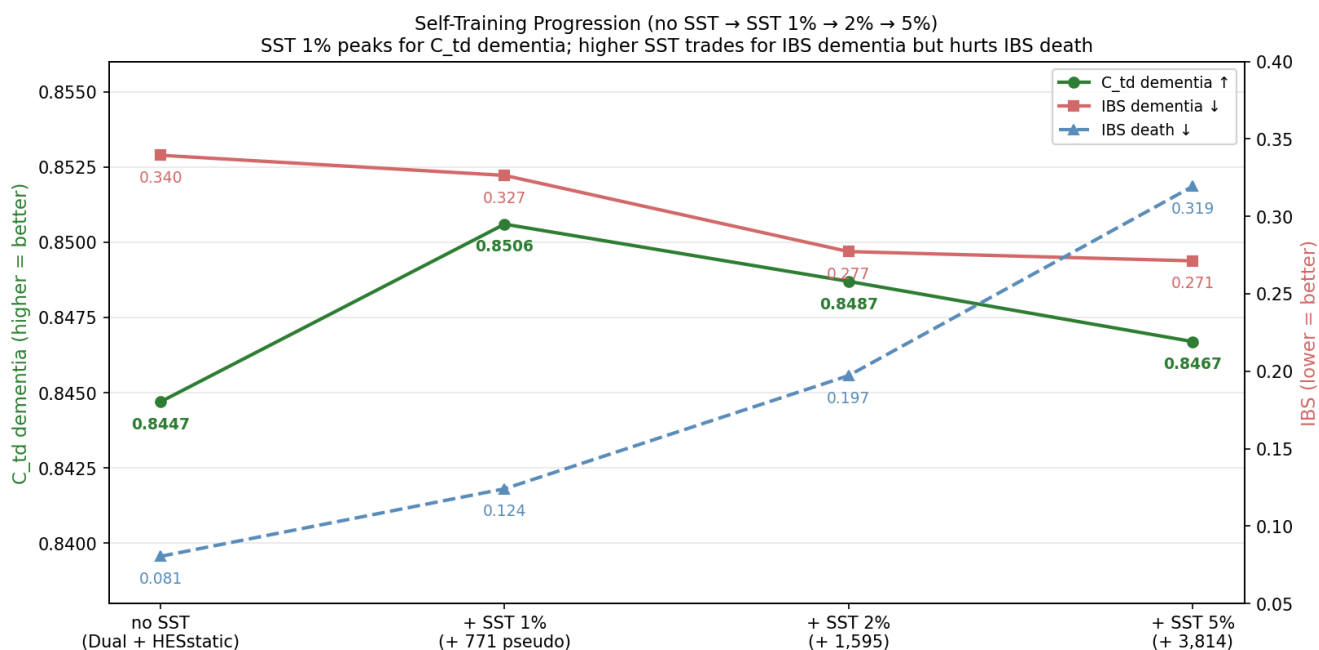
7.2 Clean-model comparison figure



The figure shows C_{td}, AUROC, and PPV@1% across the five clean-data models. Key observations:

- **Single-GP + HESstatic** $\approx$ **Dual + HESstatic** in C_{td} and AUROC — the dual HES backbone contributes essentially nothing to discrimination on top of the 22-dim HES static features.
- **SST improves C_{td}** at 1% (peak), then degrades it at 2% and 5%.
- **PPV @ top 1% drops with SST** — pseudo-labels spread the high-risk predictions wider, sacrificing top-end precision for broader recall.

7.3 SST progression



Going from Dual-gated + HESstatic → + SST1% → + SST2% → + SST5%:

- **Dementia C_{td} peaks at SST1%** (top-1% pseudo-labels), then monotonically decreases as more pseudo-positives are added
- **Dementia IBS improves monotonically** (lower = better) — more pseudo-positives → better calibration of dementia probability
- **Death IBS degrades significantly** at SST2% / SST5% — adding many pseudo-dementia cases distorts the death-vs-dementia competing-risk balance
- → **SST1% is the discrimination-calibration sweet spot**

7.4 Where do the gains come from?

Decomposing the cohort-level C_{td} gain along the project's methodological journey (V2-labels-fair comparison; full per-model numbers in Appendix B for the pre-clean models):

Step	From → To	ΔC _{td} dementia
Dual-HESaug-leak (pre-clean baseline)	—	0.8136 (start)

+ 22-dim HES static features	Dual-HESaug-leak → Dual-gated-leak	+ 0.028 ☆
+ V2 label correction (HES dementia override + DEATH cause-of-death relabel)	Dual-gated-leak → Dual + HESstatic (clean)	+ 0.003
+ 1st-round SST (top 1%)	Dual + HESstatic → Dual + HESstatic + SST1%	+ 0.006
Total improvement		+ 0.037

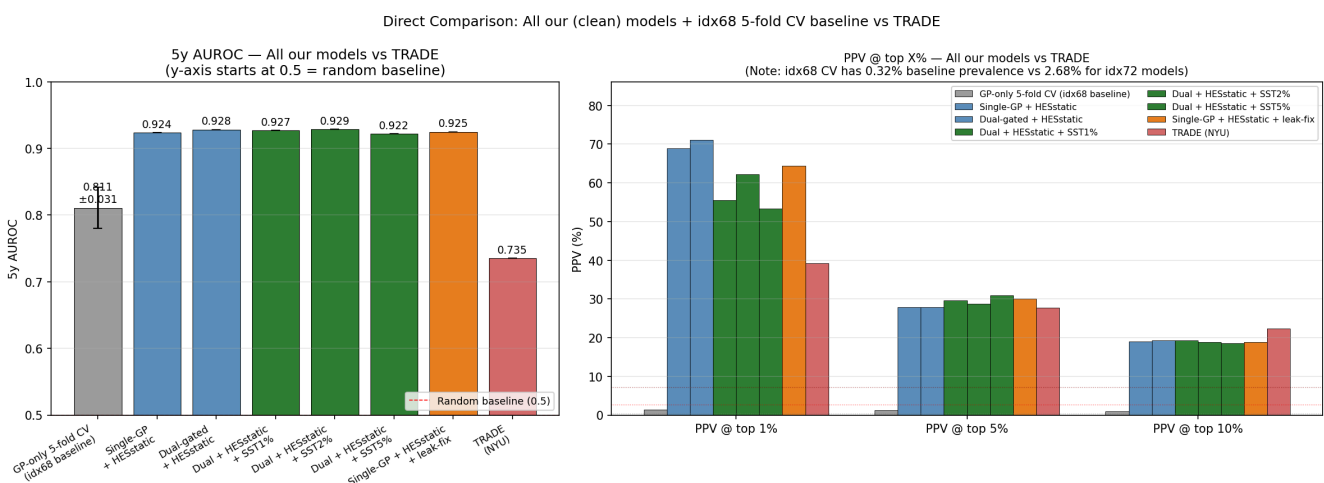
The single largest contribution is the 22-dim HES static handcrafted features (+ 0.028) — these contribute more than any subsequent step. Self-training and V2 label correction together add ~ + 0.009. The dual HES backbone (12M extra parameters) contributes essentially zero on top of the 22-dim static features (see §7.1 — Single-GP + HESstatic and Dual-gated + HESstatic have nearly identical C_{td}).

Caveat: The first two rows of the table (Dual-HESaug-leak and Dual-gated-leak) were trained on V1-era data with the pre-fix HES temporal leakage. Their cohort C_{td} values (0.8136, 0.8416) are re-evaluated against V2 labels overlaid via patient-ID match, but the **models themselves saw leaky training data**. Consequently the "+ 0.028" gain attributed to "+ 22-dim HES static" is estimated against a partially-leaky baseline. A clean re-test of this attribution would require retraining Dual-HESaug on the V2 clean dataset (listed in §10.1).

8. Comparison with TRADE

TRADE (NYU, Wisnivesky et al.) is the most directly comparable published EHR foundation model for dementia prediction. It targets a broader umbrella (AD + ADRD + MCI), uses a US EHR cohort, and reports fixed-horizon binary metrics.

8.1 Headline numbers (5-year horizon)



Metric	TRADE	Single-GP + HESstatic (ours)	Dual + HESstatic + SST1% (ours)
5y AUROC	0.735	0.9236	0.9271
5y PPV @ top 1%	39.2%	68.89%	55.56%

5y PPV @ top 5%	27.8%	27.88%	29.65%
5y PPV @ top 10%	22.4%	19.03%	19.25%
5y event prevalence (post-exclude-early-censored evaluable subset)	~7.2%	121 / 4,517 = 2.68%	121 / 4,517 = 2.68%
C-index	not reported	0.8451	0.8506

Prevalence-adjusted "lift" (PPV / prevalence):

	Single-GP + HESstatic Dual + HESstatic + SST1% TRADE		
Lift @ top 1%	25.7 ×	20.7 ×	5.4 ×
Lift @ top 5%	10.4 ×	11.1 ×	3.9 ×
Lift @ top 10%	7.1 ×	7.2 ×	3.1 ×

8.2 Side-by-side methodology comparison

	TRADE (NYU)	Ours
Backbone	BERT-style EHR FM	SurvivEHR (GPT-style)
Cohort	US, NYU Langone, ~142K patients	UK Biobank linked GP + HES, ~133K
Index policy	Sliding-window across visits (multiple samples per patient)	Cohort entry at age 72 (one sample per patient); dynamic prediction from latest pre-index event
Outcome	AD + ADRD + MCI (umbrella, ~7.2% 5y prevalence)	Dementia (AD + most ADRD, no clean MCI; ~2.68% 5y prevalence)
Output	Fixed-time binary probability	Full 25-year CIF (1,000-point grid) for two competing causes
Metrics reported	AUROC, PPV@top X%	C _{td} , IBS, AUROC, PPV@top X%, calibration (planned)
Code system	ICD-10-CM (US, 5-char)	Read v2 (GP) + ICD-10 WHO (HES, 4-char)

8.3 Code-set overlap analysis

We performed an exhaustive ICD-10 code-by-code comparison between our V2 label definition and TRADE's published code list (their Table S1).

TRADE's code list (their Table S1):

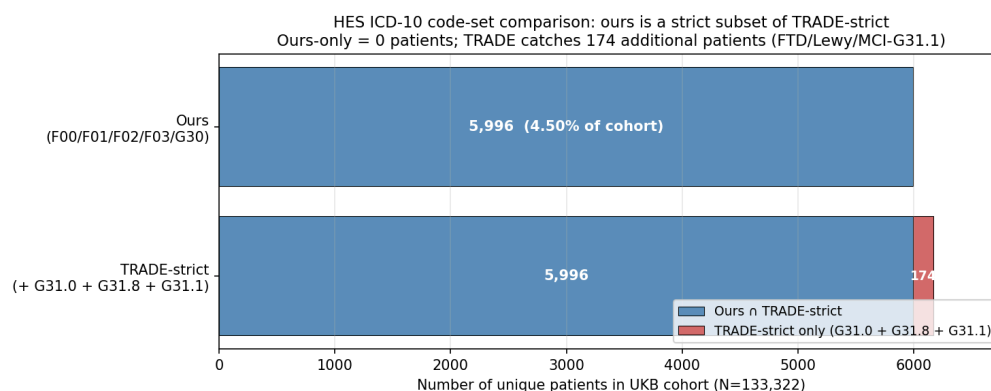
Bucket	ICD-10-CM codes (US)
AD/ADRD	F01., F02., F03., F04., G23.1, G30.*, G31.01 (Pick's), G31.09 (Other FTD), G31.83 (Lewy body), G31.9
MCI	G31.1, G31.84 (MCI), G31.85 (Corticobasal degen)
Medications	Donepezil, Galantamine, Memantine, Rivastigmine, Tacrine

Code-by-code comparison on UKB 133,322 cohort (HES side):

TRADE code	Patients in cohort (%cohort)	Our HES code set	Description
F01.*	1,417 (1.06%)	✓ Yes	Vascular dementia
F02.*	708 (0.53%)	✓ Yes	Dementia in other diseases (Pick's, Parkinson's, Huntington's, CJD)
F03.*	3,048 (2.29%)	✓ Yes	Unspecified dementia
F04.*	7 (0.005%)	✗ No	Organic amnesic syndrome — debatable as dementia
G23.1	110 (0.083%)	✗ No	Progressive supranuclear palsy — movement disorder, not pure dementia
G30.*	2,817 (2.11%)	✓ Yes	Alzheimer's disease
G31.0 (= US G31.01 + G31.09)	119 (0.089%)	✗ MISSED	Pick's disease / circumscribed brain atrophy — should be added
G31.8 (= US G31.83 + G31.85 + Lewy body etc.)	530 (0.398%)	✗ MISSED	Other degenerative — includes Lewy body dementia
G31.9	1,173 (0.880%)	✗ No	Unspecified neurodegen — debatable as dementia
G31.1	17 (0.013%)	✗ No	Senile degen NEC (TRADE's MCI bucket)
G31.84 (MCI)	0	✗ N/A	Does not exist in WHO ICD-10 — UK has no equivalent
G31.85 (Corticobasal)	folds into G31.8 (530)	✗ partial	Falls under G31.8 in WHO
Dementia meds	not included	✗ No	We do not use medication-based label triggers

Note: WHO ICD-10 (used in UK HES) has only 4-character resolution. The US ICD-10-CM 5-character extensions G31.83 (Lewy body) and G31.84 (MCI) are not representable. The closest mapping is G31.0 → US G31.01/.09 and G31.8 → US G31.83/.85.

8.4 Set overlap visualisation



Quantitative summary (133K UKB cohort, HES side):

Set	Patients	%cohort
Ours (F00 $\cup$ F01 $\cup$ F02 $\cup$ F03 $\cup$ G30)	5,996	4.50%
TRADE-strict (adds G31.0, G31.8, G31.1 — clean dementia subtypes)	6,170	4.63%
TRADE-full (adds also G31.9, F04, G23.1 — debatable)	7,026	5.27%
Intersection: ours $\cap$ TRADE	5,996	(100% of ours)
Only in ours (not in TRADE)	0	—
Only in TRADE-strict (not in ours)	174	0.13%
Only in TRADE-full	1,030	0.77%

Interpretation:

- Our HES dementia set is a **strict subset** of TRADE's umbrella (100% of our patients are in TRADE).
- TRADE catches **174 additional patients** (vs ours, strict comparison) — mostly Lewy body (G31.8, 148 unique) and Pick's/FTD (G31.0, 13 unique).
- The "debatable" TRADE addition (G31.9 unspecified neurodegen) accounts for 1,030 of the 1,030 extras under the full umbrella — 88% of the gap is from a single code that may not represent dementia per se.
- **MCI is fundamentally unrepresentable in UK ICD-10.** The 22 patients labeled `Eu057` (the closest available Read code, mapping to F06.7 "Mild cognitive disorder due to brain damage") is a different clinical construct than Petersen MCI.

8.5 If we extended our label

To match TRADE more closely we could:

- **Add G31.0 + G31.8 (clean dementia subtypes)** $\rightarrow$ +161 patients in cohort (+2.7% relative) $\rightarrow$ new label $N=6,157$ (4.62% of cohort) — essentially matching TRADE-strict (4.63%). Easy fix, would require re-running V2 dataset build.
- **Add G31.9** $\rightarrow$ +1,030 patients but introduces non-dementia noise. Not recommended.
- **Add dementia medications** (Donepezil etc., BNF-coded in GP) $\rightarrow$ could meaningfully expand to MCI-like patients on memory medications. Would require new code list curation and dataset rebuild.

8.6 AUROC / PPV computation verification

The AUROC and PPV numbers in this report were independently verified by three computations:

1. `sklearn roc_auc_score`
2. Manual Mann-Whitney U statistic via `numpy ranks`
3. Direct all-pairs enumeration (every positive–negative pair compared explicitly)

All three methods agree to 1e-16 on Dual + HESstatic + SST1%'s 5y AUROC of 0.9271. The cohort-level computation has no batching, no order-dependence, no censoring-weighting

model — methodology issues like the per-batch C_{td} bug (see Appendix A) cannot recur here.

9. Where Our Work Sits in the Literature

Brief context — these are not direct apples-to-apples comparisons because of cohort/task differences:

Paper	Cohort	Task	Approach	Reported 5y AUROC
SurvivEHR (Gadd 2025)	23M CPRD pretraining → various FFT tasks	19 multi-long-term conditions	Same backbone we use	0.79 for dementia (different prediction setup)
TRADE (Wisnivesky NYU, undated)	142K NYU Langone EHR	AD/ADRD/ MCI binary	BERT-style EHR FM	0.735
eRADAR (NYU baseline)	(same as TRADE)	AD/ADRD/ MCI	Logistic regression (hand features)	0.688
PROMISE-AD (Lyu et al. 2026, Yale)	ADNI/TADPOLE, ~1.7K	CN→MCI, MCI→AD conversion	Tabular transformer + survival	0.84 (CN→MCI), 0.997 (MCI→AD) — different paradigm
DemRisk (Reeves 2024)	UK CPRD GOLD, ~400K	5y dementia	Cox proportional hazards	0.78
Ours (Single-GP + HESstatic)	133K UKB GP + HES	Dementia 25y CIF	SurvivEHR + 22-dim HES static	0.9236
Ours (Dual + HESstatic + SST1%)	133K UKB GP + HES	Dementia 25y CIF	+ dual backbone + 1% SST	0.9271

TRADE is the only directly comparable foundation-model EHR work on a dementia umbrella; PROMISE-AD operates in a completely different paradigm (MCI→AD biomarker-driven, ADNI cohort). The numerical gap to TRADE is large (~ +0.19 AUROC) but the cohort and task differences are equally large — the **methodological message is that competing-risk modeling + dynamic prediction + foundation-model backbone yields strong discrimination on UK primary-care data.**

10. Open Questions and Next Steps

10.1 Re-run prior experiments on clean data + complete the leak-fixed SST round + idx72 GP-only CV

1. **GP-only 5-fold CV at idx age 72** (*in progress*) — replacement for the current idx68 row in §7.1, on the matching cohort. Each of 5 folds will be trained with single GP backbone, no HES at all, idx age 72, V2 clean labels. Will be the cleanest non-HES

baseline.

2. **Run the SST1% round on top of the leakage-fixed base model** — the base is already trained and tested (Single-GP + HESstatic + leak-fix in §7.1: C_{td} dem 0.8443, AUROC 0.9246, IBS dem 0.2887). Next step: run inference on the leakage-cleaned training set, select top-1% censored as pseudo-positives with empirical-lag-sampled event times, retrain. Expected to slightly improve PPV@5% and IBS dem (matches V2-Dual → SST1% trajectory).
3. **Retrain the dual-cross-attention model on V2 clean dataset** — not yet done; would isolate whether cross-attention vs gated fusion matters on clean data. The pre-leakage comparison (Dual-XAttn-leak ≈ Dual-gated-leak within 0.001) suggests no, but should be verified.
4. **Retrain Dual-HESaug on V2 clean dataset** — could match or surpass Single-GP + HESstatic since HESaug used a slightly different label augmentation strategy.
5. **Retrain GP + HESseq-fusion on V2 clean dataset** — even if it still fails on clean data, the negative result is publishable evidence that sequence fusion of two long EHR token streams is not the right strategy.

10.2 Extend the dementia code definition

Current label is 41 Read v2 codes + 5 HES ICD-10 families. Multiple extensions worth piloting:

Extension	Expected new patients in cohort	Effort	Rationale
Add G31.0 + G31.8 to HES list	+161 (+2.7%)	Low (rebuild dataset)	Clean dementia subtypes (Pick's/FTD + Lewy body) currently missed
Add dementia medications (BNF: Donepezil, Galantamine, Memantine, Rivastigmine, Tacrine)	likely +500-1500	Medium (need BNF code curation)	Many MCI / early-dementia patients are on memory medications without a formal dementia code. Matches TRADE's strategy.
Add UK MCI-proxy Read codes (1B1F . memory loss, E260 . cognitive impairment, 1S4 . . etc.)	likely +200-800	Medium (clinical curation)	Closes the MCI gap with TRADE (G31.84 has no WHO equivalent)
Add G31.9 (degenerative NS unspecified)	+1,030 (~17% relative gain)	Trivial	TRADE includes; debatable as pure dementia

The medication-based and MCI-Read-code extensions should be discussed with a clinical collaborator before adoption.

10.3 Extend the HES static covariate vector

Current 22-dim HES static is mostly cardiovascular + a few neurological / metabolic flags. Reasonable extensions:

Extension	Dimensions	Rationale
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Parkinson's disease. UmlsBERT (Michalopoulos et al., 2021, NAACL) showed that injecting UMLS Metathesaurus structure into a clinical language model improves downstream NER and NLI tasks.

"UmlsBERT integrates domain knowledge during pre-training via two strategies: (i) connecting words that have the same underlying 'concept' in UMLS, and (ii) leveraging semantic type knowledge in UMLS to create clinically meaningful input embeddings."

Two integration paths for our SurvivEHR backbone:

1. **Token-level concept embedding** — for each Read v2 / ICD-10 code in our vocabulary, look up its UMLS CUI (concept unique identifier) and semantic type. Add a learned embedding for the CUI and/or semantic type to each token's input embedding. This injects structured "what kind of clinical thing is this" knowledge.
2. **Knowledge-augmented continued pretraining** — continue pretraining the SurvivEHR backbone with an auxiliary objective that pulls codes with the same UMLS concept closer in embedding space. Could use a MASK-then-predict-CUI task analogous to UmlsBERT's masked LM with concept distribution.

Expected benefits:

- Better few-shot generalization to rare codes (most of our 41 dementia codes have <10 patients)
- Better cross-vocabulary alignment between Read v2 (GP) and ICD-10 (HES) — currently the model has to learn this from co-occurrence statistics
- Improved interpretability — embeddings would cluster by semantic type

Implementation cost is moderate (UMLS license is free for research; CUI lookup is straightforward via QuickUMLS or RxNorm).

10.7 Methodological improvements

1. **IPCW-AUC** (Inverse Probability of Censoring Weighting) as a principled alternative to the "exclude early-censored" policy — most rigorous censoring-aware AUC estimator. Recommended as the headline metric for the final paper.
2. **Report time-dependent AUC** at horizons 1, 2, 3, 5, 10y to show how discrimination changes with horizon.
3. **External validation cohort** — currently all evaluation is in-sample on UKB. A held-out cohort like CPRD GOLD or US Optum would strengthen the paper substantially.

Appendix A: Methodology corrections (project history)

These corrections are not central to the headline narrative but were important learnings.

A.1 Per-batch C_{td} averaging artifact

The default Lightning training callback aggregates C_{td} **per batch** then averages across batches. This is not a bug in the strict sense — it's the callback's default aggregation — but it

produces values that are not equivalent to cohort-level C_{td} . With batch size 16:

- 47% of batches contained 0 dementia events, making per-batch C_{td} numerically unstable.
- Cross-batch (positive, negative) ranking pairs were lost.
- Result was order-dependent: sequential vs shuffled batching changed mean C_{td} by +0.078.
- Reported C_{td} values were systematically **understated by ~ 0.07 - 0.10 absolute** (e.g., Dual + HESstatic + SST1% per-batch = 0.7685 vs cohort-level = 0.8506).

Fix: Compute C_{td} at the cohort level using

`pycox.EvalSurv.concordance_td('antolini')` on the full test set in a single pass (script: `compute_cohort_ctd_generic.py`; see `COHORT_CTD_RESULTS.md` for the authoritative numbers used in §7.1). Cross-verified with a manual numpy implementation (Wolbers 2014 competing-risk extension); cause-specific and true-competing-risk C_{td} differ by < 0.005 on our data. All numbers in §7 are cohort-level from this script.

A.2 GP-prevalent label leakage

In the V1 dataset, **246 test/train/val patients had a GP dementia code recorded before their cohort entry date** (idx age 72) but were still included as candidates for the dementia event. This is a temporal leakage: they were prevalent cases, not incident cases.

Fix: The latest "Single-GP + HESstatic + leak-fix" model uses a rebuilt base dataset that removes these 246 patients (213 train / 17 val / 16 test). Test-set numbers in §7 already exclude the 16 test-set leaky patients (canonical 8,241 cohort).

A.3 25-year horizon clarification

We initially believed the model predicted a 5-year horizon. After investigating the empirical pseudo-event lag distribution (median 8.8y, max 16y for true V1-Type-C alive-censored patients), we discovered the actual model horizon is 25 years: $\text{time_scale} = 1825 \text{ days} \times \text{supervised_time_scale} = 5 = 9,125 \text{ days} \approx 25 \text{ years}$. The normalised $t_{\text{eval}} \in [0, 1]$ therefore maps to $[0, 25 \text{ years}]$. All metrics in §7 are computed against this correct horizon, with $t_{\text{eval}}[200] = 0.2002 \approx 5.005y$ used for "5y" results.

A.4 Pre-leakage era HES temporal filtering

In V1 datasets, HES records were not strictly filtered to `admidate < index_date`. This meant patients whose dementia diagnosis was recorded in HES after age 72 could still have features influenced by post-index admissions — a form of temporal data leakage.

Fix: The V2 clean dataset rebuilds the 22-dim HES static feature vector with strict pre-index filtering (`admidate < index_date`, plus dementia codes excluded). V2 clean labels also use cleaner cause-of-death coding. The pre-leakage experiments (Dual-HESaug, GP + HESseq-fusion, Dual-gated, Dual-XAttn) were trained before this fix. The GP-only 5-fold CV baseline (idx age 68) does not have HES temporal leakage (uses no HES data) but is on a different cohort — see §7.1.

Appendix B: Pre-clean (leakage-era) experiments

The following experiments were trained on the V1 dataset, where pre-index HES dementia codes were not always strictly excluded from the static feature vector (temporal leakage), and labels did not include the V2 corrections. We re-evaluated them at the cohort level with V2 labels overlaid via patient-ID match to enable fair comparison; their numbers should not be treated as headline results.

Short name	Architecture
Dual-HESaug-leak	Single GP backbone + HES label augmentation + 22-dim HES static
GP + HESseq-fusion-leak	Single backbone with HES events fused INTO the GP token sequence (failed approach — sequence becomes too long and 60% gets truncated)
Dual-gated-leak	Dual GP + HES backbones, gated fusion, + 22-dim HES static
Dual-XAttn-leak	Dual GP + HES backbones, cross-attention fusion, + 22-dim HES static

Cohort-level C_{td} (dementia) on V2 canonical 8,241 test cohort, V2 labels overlaid:

Model	Cohort tested on	Cohort C _{td} dem	Note
Dual-HESaug-leak	V2 canonical 8,241 (V2 labels overlay)	0.8136	fair vs clean models
GP + HESseq-fusion-leak	V2 canonical 8,241	0.7098	sequence-fusion approach fails: – 0.10 vs Dual-HESaug-leak
Dual-gated-leak	V2 canonical (V2 labels)	0.8416	dual backbone trained on V1 labels
Dual-XAttn-leak	V2 canonical (V2 labels)	0.8428	cross-attn ≈ gated

Key methodological lessons from these pre-clean experiments:

- **Sequence fusion of GP + HES into one token stream fails** — the combined sequence median is ~731 tokens vs the model's block size 512, so 60% of patients get truncated. Late fusion (dual backbone or static features) is strictly better.
- **Gated fusion ≈ cross-attention** within 0.001 C_{td} — the fusion mechanism is not where the gains live.
- **Dual backbone ≈ single backbone + 22-dim HES static** within 0.001 — the dual model's second backbone is essentially redundant once the 22-dim static features are present (later verified on clean V2 data).

Appendix C: Key files

Item	Path
Project plan	CPRD/PROJECT_KNOWLEDGE.md

Cohort C _{td} results (all models)	CPRD/COHORT_CTD_RESULTS.md
Methodology audit	CPRD/CTD_CORRECTION_AUDIT_2026-05-14.md
Cohort C _{td} generic computation	CPRD/compute_cohort_ctd_generic.py
5y AUROC + PPV computation	CPRD/compute_trade_comparison.py, compute_baseline_trade_metrics.py
3-method AUROC/ PPV verification	CPRD/verify_trade_metrics.py
Code overlap with TRADE	CPRD/compute_trade_overlap_full.py, compute_trade_missed_finegrain.py
Leakage-fixed dataset build script	CPRD/v6_step2_build_v6_base_dataset.py
In-progress training config	CPRD/examples/modelling/SurvivEHR/confs/ config_FineTune_Dementia_CR_v6_base.yaml
22-dim HES feature builder	CPRD/examples/modelling/SurvivEHR/ build_hes_summary_features.py
Figures in this report	CPRD/figs/fig_*.png